Verification Plan

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Plan A is predicated on the ability to enforce an international deal to slow down AI progress without relying on trust. We are confident that verification measures can make this possible, [1] by providing sufficient assurance to both the US and China that defection by the other would either be caught or small enough to be inconsequential. [2]

There is a spectrum of how costly and how trustworthy our best verification solutions are, which we can think of as a verification frontier. Today, the verification frontier is not mature. Currently the only way of verifying an AI development pause with extremely high assurance between the US and China would probably require datacenters to be powered down, [3] and anything less costly than this might have significant risk of being violated. [4] That being said, we believe the verification problem is inherently defense dominant, [5] and AI verification is also an extremely young and so far tiny field, so it should be highly tractable to make significant progress in the next few years.

This supplement provides a summary of the verification plan that we think should be followed to make Plan A viable, and then is followed by a more detailed walkthrough of how the verification story unfolds in our specific Plan A scenario.

Want to help build this? See the Get Involved tab: what needs to be done, who the field needs, an expression of interest form, and open opportunities across the verification field.

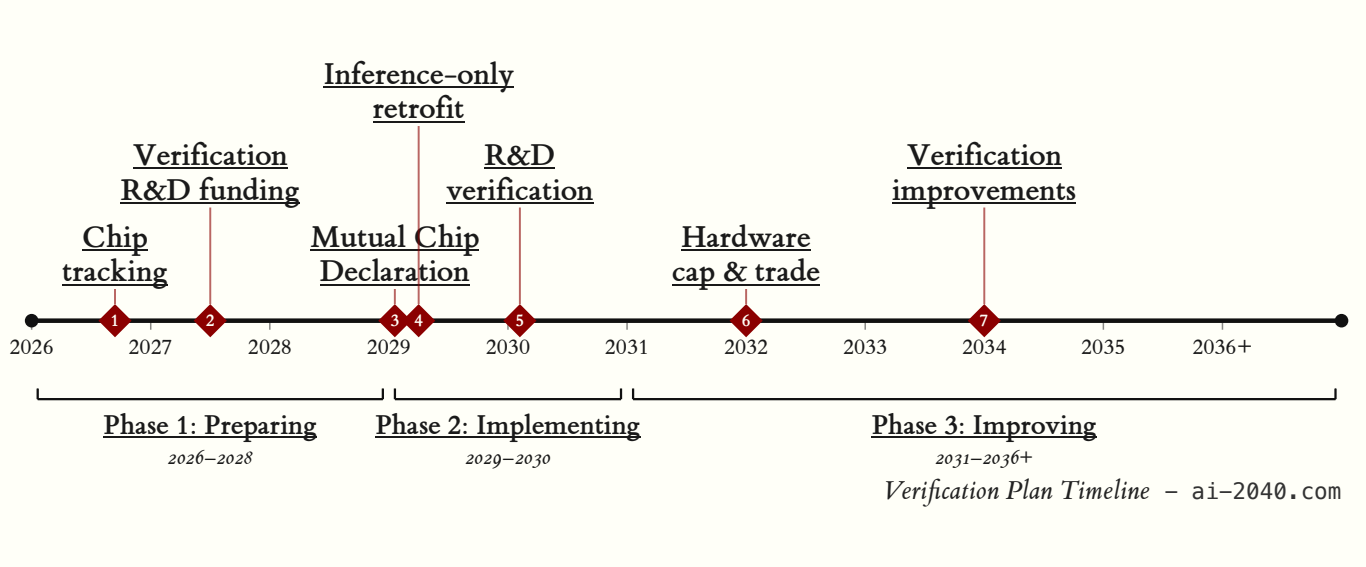
Summary of the Plan

→ Prepare deal optionality

→ Implement the deal

→ Improve robustness

Verifying the deal in our Plan A scenario



Verification Plan Timeline – ai-2040.com

Appendix: Packet-Based Verification

Summary of the Plan

We recommend an international deal to avoid a catastrophic race to superintelligence, spearheaded by the US and China. This deal needs to be enforceable without relying on trust, because the US and China will be unlikely to make an agreement that cannot be verified; and they are much less likely to defect on the agreement if any defection is likely to be caught.

We can break up the verification problem into two key components:

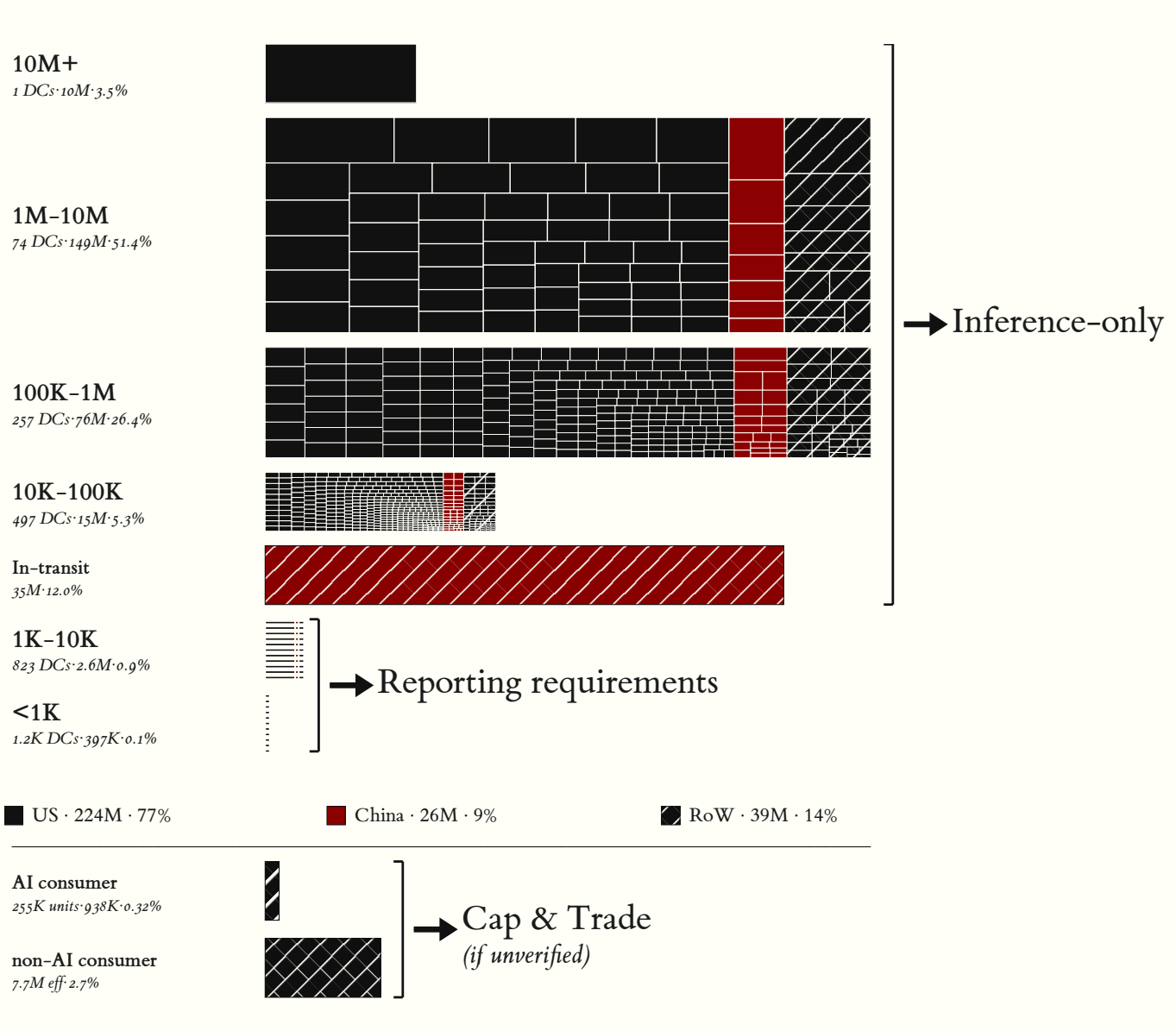
> **Problem 1. Verifying that known compute is compliant.**
>
> We are confident that it is possible to have verification measures that enable a trustless deal. That being said, if a deal needed to happen tomorrow, we would not be ready to enforce it trustlessly in a low-cost way. [6] Given the increasingly large economic incentive to keep AI services online, a ready-to-go verification solution (that doesn't require much time to activate) may be incredibly important for making a deal happen.

> **Problem 2. Verifying that unknown compute is inconsequential.**
>
> Our covert project supplement estimates that even a competently-executed covert diversion effort would net a median of about 0.5% of the world's AI-relevant compute (80% CI: 0.1% to 1.4%) as 'dark compute' at the start of the deal, [7] which given the likelihood of compute-dependence of frontier progress, should buy the deal a lot of time and stability. [8] This bound should also tighten as the deal goes on, as detection efforts accumulate and more compute is brought under verification.

Our recommended solutions come in three phases.

Phase 1. Prepare Deal Optionality. The US and China take actions to make a deal more likely to be enforceable and durable, by preparing the ability to do inference-only verification, and keeping close tabs on how much AI compute exists and who owns it.

Phase 2. Implement a Deal. The US and China perform a mutual compute declaration (spreading to third countries and ideally reaching >99% of world compute); inference-only verification retrofit to ~99% of world AI datacenter compute, and then sprint towards setting up secure R&D verification that will use new post-deal compute production.



This is what we think implementing the deal would look like in 2029 in our scenario, with an inference-only retrofit of all the medium and large AI datacenters (>10K H100e, or approx. >$100M), and this being enough to cover ~99% of world AI-relevant compute. Then keeping tabs on the rest of the smaller clusters and taking measures to avoid them being possibly used in a covert project. We are not confident in the modelling of this concentration in datacenter sizes, so the exact cutoffs and interventions may need to be different.

Compute locations by datacenter size (Jan 1 2029) – ai-2040.com

Phase 3. Improve robustness. Over time the US and China improve the stability and durability of the verification regime, especially through hardware security, verification robustness, and more.

Phase 1. Prepare Deal Optionality

We recommend focusing pre-deal effort on sprinting towards a minimal verification package that can be used to securely retrofit 99% of the world's AI compute to be verified as inference-only, and for at least 99% of the world's AI compute to be known-to-exist (at least to intelligence agencies) with low uncertainty, so that it can't be hidden from a future compute declaration (and therefore verification regime) with plausible deniability. [9]

▶ *Why do we recommend this?*

Top priority recommendations:

> **Invest in verification R&D with a specific focus on developing a secure inference-only retrofitting plan with around 80% of total verification R&D effort.**
>
> Also, support supplementary R&D efforts with around 20% of verification R&D effort pursuing promising additional measures, such as software-only solutions, privacy-preserving verification measures, and more general verification solutions (e.g., that verify the compliance of training and experiments as well).

> **Invest in compute accounting and tracking (e.g., carried out by intelligence agencies) to know how much AI relevant compute exists in the world, and which entities are in possession of it.**
>
> Notably, our recommendation for this early chip tracking doesn't prioritize knowing where this compute is, or what it is being used for. We believe that the majority of the upside in terms of making a deal viable is simply in knowing how much exists and keeping track of which actors own it at any given time, because the lack of plausible deniability should be sufficient to force this compute to be declared at the time of deal negotiation (given that there will be large incentives in favour of joining a deal). That being said, also knowing its location is helpful in avoiding cases of plausible deniability (e.g., faked records of on-selling or chip deaths) and cases where (for some reason) the US and Chinese governments have little leverage over the owning entity (and therefore can't pressure them to declare the location of the compute they are known to have, e.g., Russian compute).

Phase 2. Implement the deal

To set up a stable deal, we recommend immediately carrying out a mutual compute declaration and supply chain records audit to maximize the amount of compute that is brought under the deal. Then implementing the best available inference-only verification solution to this compute to enforce a brief pause on training and experiments. In parallel, we recommend a sprint to set up a verification regime for resumed training and experiments.

▶ *Why do we recommend this?*

Top priority recommendations:

> **Carry out a bilateral compute declaration and supply chain audit, for both existing compute and new compute production, and extend to third nations with significant compute ownership or production.**
>
> Every major datacenter owner and companies with a major role in the supply chain publicly declare their records about purchases and sales, and allow auditors and inspectors into each other's countries and companies to verify the records and count of final chips. This needs to also including compute in transit and active compute production. From this point forward, existing compute and new compute production is closely tracked and guarded so that it can't be redirected to a covert project without detection. Extend this declaration to other countries with significant compute ownership or production capacity. Do so urgently unless there is confident intelligence in place that would detect this compute being moved.

> **Implement inference-only verification on existing compute and any new compute coming online.**
>
> Roll out inference-only verification to existing compute to enforce the brief AI R&D pause while allowing AI services to remain online. Ideally there would be a highly secure solution ready to go. If not, implement the best solution available. If there is no solution at all ready then either shut down the compute, or rely only on non-technical measures (e.g., embedded auditors in each other's companies checking that no one is implementing experiments or training runs) in the meantime while sprinting to transition to a better solution.

> **Sprint to set up R&D verification in accordance with the chosen transparency regime.**
>
> We recommend a specific regime we call Total Research Transparency in the Plan A scenario, but as explained in our transparency supplement, there are conditions under which a different approach may be better. Regardless of the regime, there should be some combination of highly secure R&D clusters and highly secure post-deal inference clusters built, because across most transparency regimes we imagine at least AI model weights and verification correctness requiring SL5 (robust to nation state actors) level security. Because of this we think these clusters should be built from scratch.

▶ *What AI R&D agreements should actually be verified?*

Phase 3. Improve Robustness

Once a deal has been set up, we expect many improvements to its robustness will be possible. We recommend investing heavily in hardware security; improving the verification regime; pursuing favorable trade-offs in hardware design that favor deal robustness (e.g., by making verification easier, improving various security properties, differentially favoring the actors in the deal over covert projects, etc.); and reducing incentives to pull out of the deal (e.g., through mutually assured compute destruction, and balancing deal dissolution outcomes between actors).

▶ *Why do we recommend this?*

▶ *Concrete ideas for deal robustness*

Verifying the deal in our Plan A scenario

In our Plan A scenario, we follow the high level verification plan sketched out in the previous section. This part of the verification supplement will walk through the implementation and verification story with more concrete detail. Because it is a scenario about the future, in a very young field with ongoing R&D, we are not confident in the specific details. That being said, we hope to establish a useful concrete baseline for how the verification plan we sketched out above could actually be implemented.

2026–2028: Preparing deal optionality

2026: Chip tracking begins

The intelligence agencies in the US begin internally keeping tabs on AI compute smuggling into China, and their domestic production levels, reaching a 10% uncertainty over how much AI compute Chinese entities own in total by the end of the year. They don't do anything with this information yet (i.e., they don't share smuggling information with the BIS). China learns that the US intelligence community is doing this, and begins doing it as well by leveraging existing assets mostly out of TSMC in Taiwan to get accurate counts and records of total production. They reach 10% uncertainty on US total AI compute, mostly due to uncertainty over pre-existing compute and production outside of Taiwan (e.g., South Korea and the US). Uncertainty over new compute production is much lower (on the order of 1%).

2027: Verification R&D gets funding

The US executive branch starts expressing interest in deal optionality and verification measures that could enable this. After a series of RFIs they choose to focus efforts on a network-tap and recomputation centered approach for an inference-only retrofitting solution. The prospect of installing devices in each other's datacenters raises initial alarm bells but it's a tractable security problem to work on. Ultimately due to the sheer simplicity of some of the possible network tap designs (essentially a piece of glass splitting optical signals), and parallel investment into various physical security agendas, it becomes a palpable possibility that these taps might actually be worth producing en-masse in case they were to be needed. Unilaterally trusted recomputation servers (that the network taps redirect traffic to and are responsible for actually performing the verification) are more complicated and thus harder to trust by default, but there are large investments made here. The R&D programs are purposely transparent and open sourced to build trust. China also sees the incentive to develop and build their own symmetrical solution in a similarly transparent way. In the latter half of the year, they agree to mutual information sharing about important verification-relevant information which is otherwise not sensitive, especially about each other's networking and computing paradigms, which is necessary in order to build compatible network taps and recomputation servers. [13] Several pilots from different sources of funding and different places around the world are carried out over the course of the year, improvements to the designs are iterated on, and preparations are made to increase production scales. Ultimately the US and China end up opting for network tap solutions with extremely simple optical splitting designs that have minimal room for any vulnerabilities, rather than some of the more ambitious designs that we propose in the box below, or different cryptographic based-approaches, which are more uncertain, but if they work (at sufficiently low cost) they might be even better and easier to trust.

▶ *Concrete inference-only retrofitting proposal.*

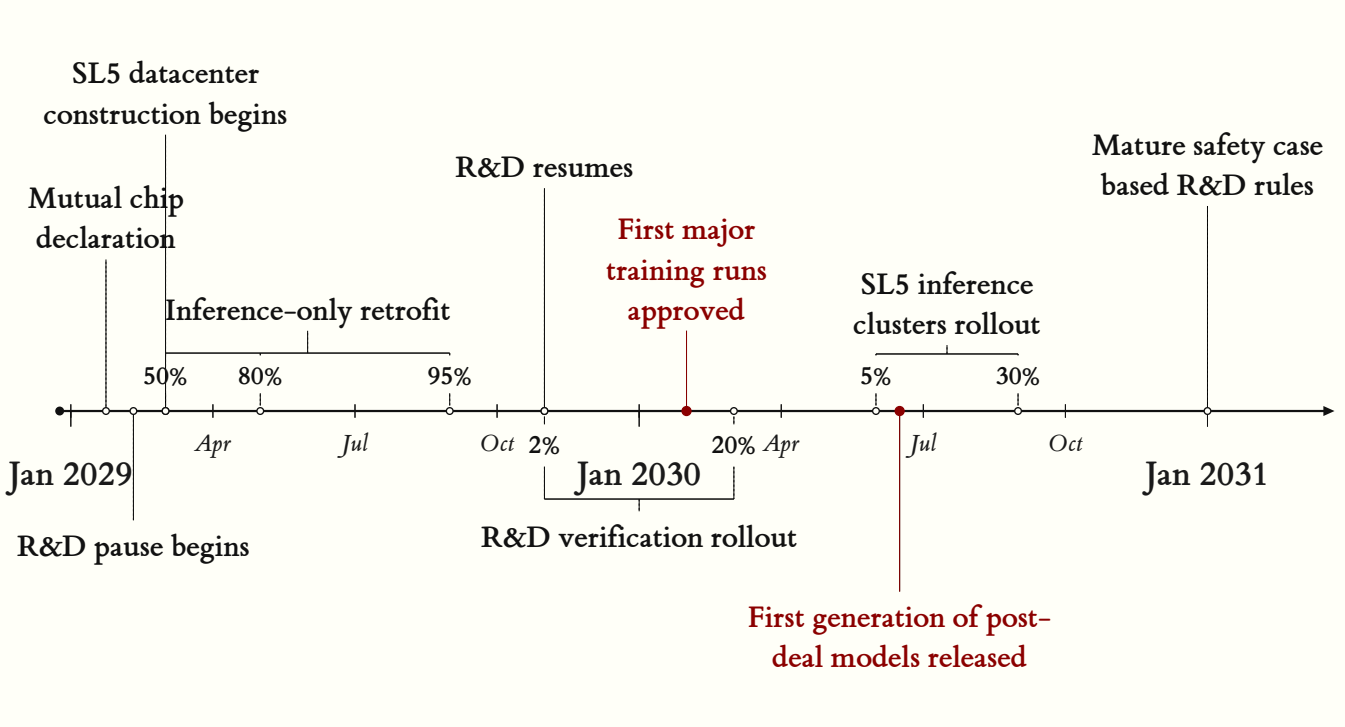
On top of this key effort to build the network-tap and recomputation centered approach, there is also significant funding (order of $100M) mobilized into R&D for other approaches, including software-only and cryptography-centric (e.g., ZKPs) approaches, and a similar pool for physical security R&D (order of $100M) that supplements the inference-only retrofitting solution.

▶ *Other promising verification directions.*

2028: Chip tracking is mature

The intelligence agencies have been keeping close track of new compute production, and have been partially successful in narrowing down their uncertainty over how much AI compute pre-dated their tracking efforts. By the end of the year, China has around 2% uncertainty over the US' total compute, and the US has around 4% uncertainty over China's. This means from China's perspective, they are 80% confident that the US owns somewhere between 222M and 226M H100-equivalents. From the US' perspective, they are 80% confident that China has somewhere between 25M and 27M H100-equivalents.

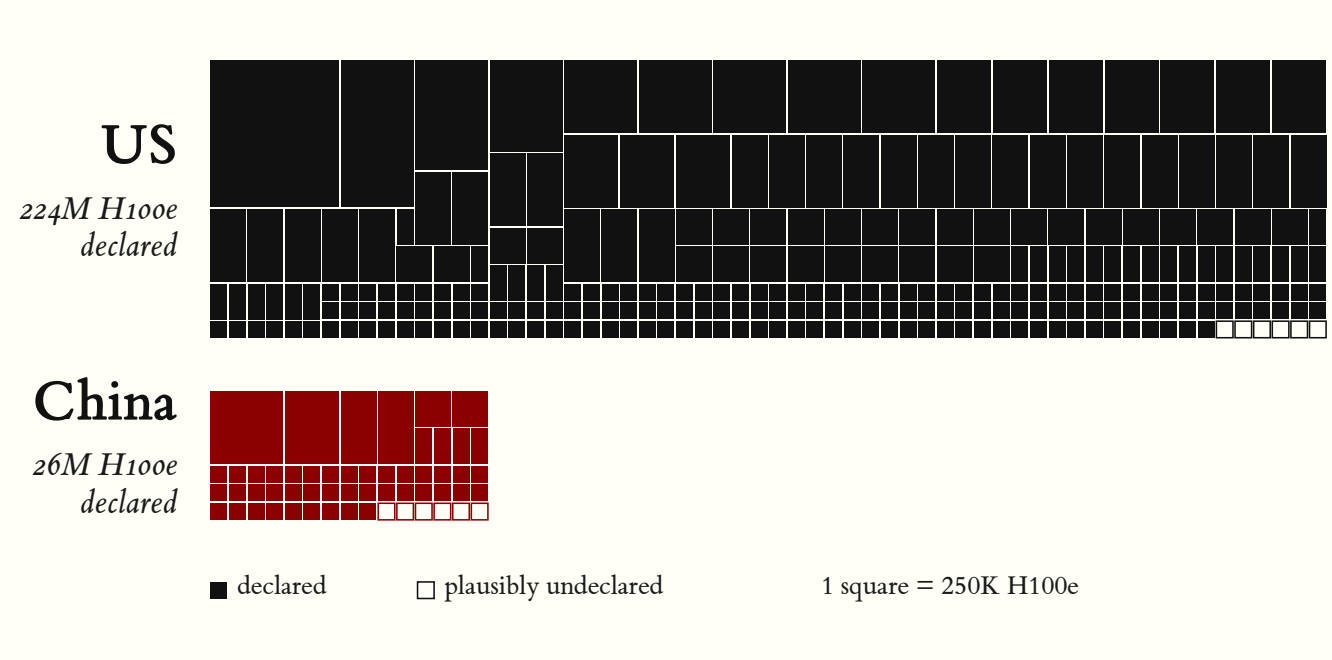
2029–2030: Deal Implementation


SL5 datacenter construction begins
Mutual chip declaration
R&D resumes
First major training runs approved
Mature safety case based R&D rules
SL5 inference clusters rollout
Inference-only retrofit
50%
80%
95%
5%
30%
Apr
Jul
Oct
2%
20%
Jan 2029
Jan 2030
Jan 2031
R&D pause begins
R&D verification rollout
First generation of post-deal models released

Deal Implementation Timeline (detailed) – ai-2040.com

Jan 2029: Mutual Chip Declaration and Inspection

Within a week of starting Plan A negotiations, the US and China have carried out a compute declaration and supply chain record audit, allowing each other to send a few hundred people into each other's countries to inspect and audit records. This lets them reduce their already relatively small uncertainties over each other's compute. Each side's best guess is now that the other could have hidden at most around 1.5M H100e from the declaration (with the 80% CI reaching about 4M), which is about 4% of the leading AI company's R&D compute and about 0.5% of total world compute. For our estimates and reasoning on how much compute a covert project could probably hide see the covert projects supplement.



Chip declaration result – ai-2040.com

More detail on the suggested interventions by cluster size bucket below:

Cluster size	Immediate intervention	Estimated compute in this category in Jan 2029 as % of global AI datacenter compute
Medium and Large datacenters (>10K H100e, ~$100M)	Inference-only retrofit	99%
Small datacenters and diffused AI compute	Reporting requirements for anything above an A100 in AI performance. Random spot checks.	1%
Non-AI compute (e.g., iPhones) with unclear AI usefulness	Intelligence to look for efforts to gather up massive quantities of consumer compute.	0.1-10%? Depends on uncertain AI-usefulness adjustments.

The chip declaration also involves keeping extremely close tabs on all new compute production, and any chips above certain performance metrics of AI-relevance need to either be directed to known locations to be included under the verification measures. While lower powered consumer compute production can likely remain unrestricted.

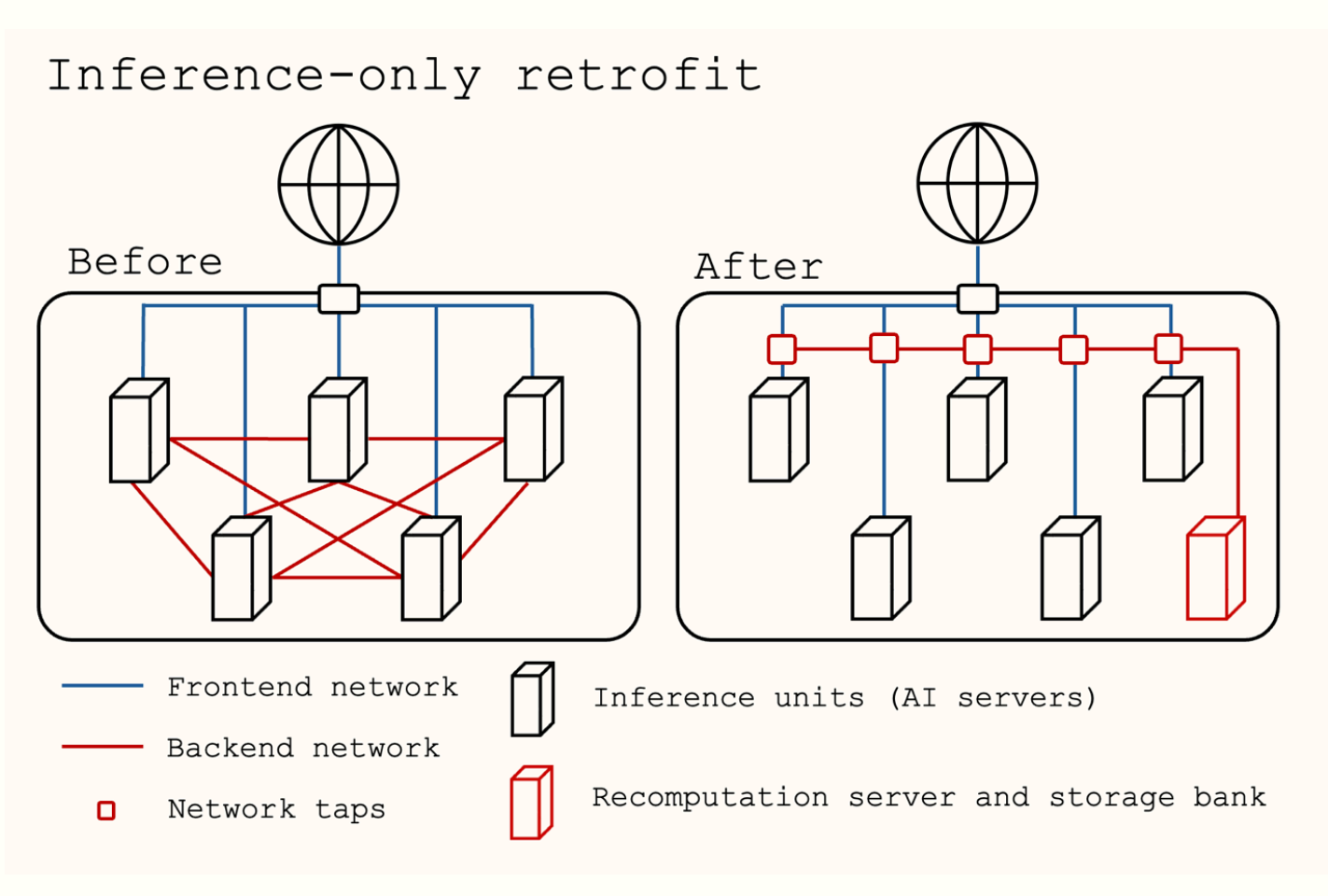
▶ *Chip flow restrictions.*

Feb 2029: Inference-only retrofit begins

After the US and China have carried out the compute declaration and inspection, and are relatively confident that they know about the vast majority of each other's compute, they handshake on a brief AI R&D moratorium and announce it to the world. AI companies are supposed to stop doing R&D immediately for now, including pausing in-process training runs. While they roll out the inference-only verification solution to each other's largest datacenters, they agree to a 100% power-off of small clusters (anything under 10K H100e, because it will take a while to get to retrofitting these) and a 80% power-off of all larger datacenters because only around 20% of the compute should be sufficient to keep all AI products and services running properly, simply at slightly reduced rates. This limits the extent to which anyone can defect and make differential gains while the deal is being set up and then just going back to racing from a better position than they were in before.

The US has about 95% of its compute in around 200,000 total AI server racks (with just over 1K H100e per server rack on average), and China has about 95% of its compute in 100,000 AI server racks (with about 250 H100e per server rack on average).

Luckily the US has a stock of around 10,000 network taps and 100 (unilaterally) trusted recomputation servers. China has a stock of around 20,000 taps and 200 servers, and they can each purchase another 10,000 per month from various third parties. As a result of China having less compute, the US can retrofit China's compute faster, but they agree to a proportional retrofitting schedule determined by China's achievable pace, which is retrofitting 40% of the US compute into inference units of 4 server racks (or 4K H100e) with their existing stock of network taps in 1 month. So they will both follow this pace, and then ramp up to 80% by the 3 month mark and then to >95% and/or smaller inference-units beyond that.

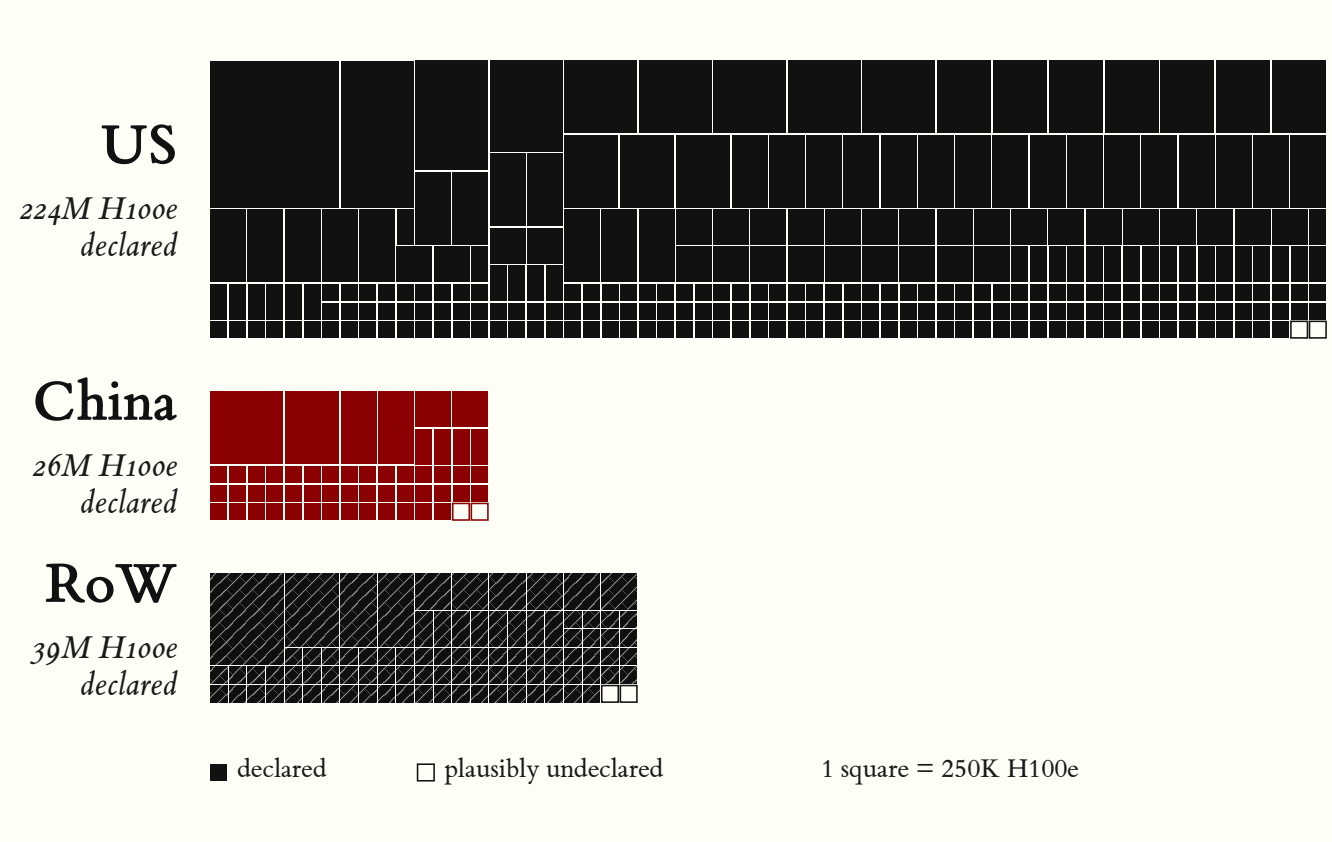


March 2029: R&D cluster construction begins

While the inference-only retrofit is being carried out, the US and China then agree on the high level regulatory plan and transparency regime they will be aiming for (some of the options are highlighted in our transparency supplement). Given the relatively small perceived risk from illegal projects and various other considerations, they decide to aim for the Total Research Transparency approach. This involves setting up R&D datacenters with SL5 model weights security and SL5 verification integrity. They want these to not be defendable so that it's difficult to try to pull out of the deal and start racing again using these new R&D clusters, so they aim for mutual deterrence by locating the US R&D clusters in Mongolia and the Chinese ones in Canada. The US has guards at their Mongolia cluster that have robust 'scorched earth' mechanisms to destroy the compute if China or someone else tries to seize it, and symmetrical measures are in place for the Chinese compute in Canada. They get started on these infrastructure projects immediately in March with high urgency, aiming to have a small amount of capacity online within 6 months so that a verified R&D pilot can start. The verification approach they plan to follow is similar to the inference verification approach, with network taps being used as evidence collection, and partial recomputation being used to verify correctness of the packets that make up the workload.

April – May 2029: Third party countries begin joining the deal

Starting in April and through the rest of 2029, most of the richest 20 countries in the world, and important semiconductor supply chain countries join the deal by doing a compute declaration and allowing the inference-only retrofitting to their datacenters, including: Germany, Japan, UK, India, France, Brazil, Canada, Mexico, Italy, Australia, South Korea, Spain, UAE, Saudi Arabia, Taiwan, Netherlands, Switzerland.



Chip declaration result – ai-2040.com

June – July 2029: Construction of new highly secure (SL5) inference-only clusters begins

As part of the transparency regime, the plan is to do broad public deployment of approved frontier models, but under SL5 model weights security achieved through SL5 output verification and distillation mitigations. The plan is to offer a mixture of Zero Data Retention (ZDR) inference for consumers, and transparent inference for higher-risk deployments. More on this in our transparency supplement. We think these clusters probably need to be built from scratch to reach SL5 model weights security and verification integrity.

▶ *Zero Data Retention (ZDR) inference*

August – September 2029: Inference-only ready on 95% of world compute

With several of the largest compute owners joining the deal over the summer, 95% of the compute in the world is now under inference-only verification, and it has now scaled to higher granularity (smaller inference units) and improved physical security (side channel reductions and high effort physical guarding and monitoring).

Late 2029: Pilot R&D verification begins coming online

In both Canada and Mongolia, mutually constructed datacenters come online at around 1GW in Canada and 2GW scale in Mongolia by September (around 2% of world compute) scaling to around 15% of global compute by the end of the year. Companies are allowed to start hosting compute in these clusters and bid over space as it comes online. Initially, only very small experiments get approved and there is red teaming and iteration on the verification and security measures to test and improve their efficacy.

▶ *An overview of possible verification approaches.*

▶ *Key ingredients to any verification approach that we will assume are pursued.*

Early 2030: Full R&D verification underway

There are now around 100M H100-equivalents in the verified R&D clusters in either Canada or Mongolia (which is about 20% of world compute at the time and equal to 35% of the compute that existed at the start of the deal). These clusters have the transparency measures in place and now companies are approved to do the first major post-deal training runs using combined algorithms. In a few months these training runs complete and several of the models are around the automated coder milestone (AC). The models get broadly tested and red-teamed, including by many third party groups and auditors that rent small amounts of compute in the R&D clusters for this purpose.

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[1] With variable economic costliness and other tradeoffs which we discuss. As a baseline we are >90% confident that a highly costly verification solution is possible for both inference and training (e.g., multi-year verified compute shutdown with verification R&D sprint and secure rebuild).

[2] Our covert project supplement discusses how likely it is for a covert project to be able to pose a threat to the deal or win the AI race.

[3] Even this might be difficult to execute on to the extent that at least one side has high uncertainty over the other having a large secret datacenter that they don't declare and allow the other side to verify is powered off.

[4] Another live option that could currently be executed quickly would be to unplug (most) backend networking from datacenters. This might allow a lot of inference to continue (less costly), but also might not slow down AI development robustly (inference is used as rollouts for data to be used in federated learning training run elsewhere).

[5] A key reason we believe this is that the verifier only needs to catch one violation, while a prover needs to successfully hide all of their covert computations. This is the reason behind e.g., the favourable scaling of partial recomputation due to random sampling which we explore in our appendix.

[6] I.e., doesn't require compute to be shut down for long.

[7] Conditional on the Plan A scenario (the mutual chip declaration and inspection in 2029 carried out, and our chip tracking recommendation executed competently), our covert project supplement estimates that a competently-executed PRC diversion effort would net a median of 0.51% of world compute (80% CI: 0.11% to 1.4%), implying roughly an 80% chance that total 'dark compute' stays under 1% of the world's AI-relevant compute.

[8] We are uncertain about the compute-sensitivity of progress in the takeoff period (automated coder to superintelligence milestones). If we operationalize this (for simplicity) in terms of how much slower you go if you have 10x less compute, we think it's most likely to be somewhere between 2x to 6x slower (see How much slower does takeoff go with 10x less compute? and the matching model in the covert project supplement). The specific trajectories we expect hypothetical covert projects to follow in the Plan A scenario (including the effects of algorithmic progress leaking from legal projects) are modelled and explained much more carefully in our covert project supplement.

[9] By known-to-exist, we are recommending that intelligence agencies know with high confidence at least that the compute exists, and who has it, not necessarily exactly where it is (though that is also helpful), because the first two conditions should be sufficient to make it impossible to exclude from a future mutual chip declaration with plausible deniability.

[13] For example, technical details about the networking, e.g., optical speeds, bandwidths, frequencies.

There is now a big regulatory question of how to control the speed of R&D progress to balance the risks (of going too quickly) and benefits (of improving capabilities). This will arguably be the hardest part of the deal to execute on competently. There is a complex set of dependencies, where going too fast incurs AI takeover risk and/or causes progress to diffuse to potential covert projects (reducing the slack the legal project has), while going too slow incurs risk of making less progress on alignment before the deal possibly breaks down, or indeed actually losing to an illegal project operating outside of the deal.

The high level capability schedule we recommend is to scale to the maximum capability you can reliably make control-based safety-cases for, ideally mostly with hardware scaling, and then pause until your marginal progress on alignment-based safety-cases (handoff risk) is lower than the risk being incurred by the deal.[17]

But even with a desired high level capability schedule in mind it may be hard to actually enforce this level of capabilities progress, especially to the extent that capabilities progress is unpredictable from inputs (e.g., compute), hard to measure, or fast paced by default. Our best guess is that harsh compute caps, and then a mixture of quality ad-hoc rules will be sufficient, as explained in more detail in the below box.

▶ *Research Titration: How to control the speed of R&D progress?*

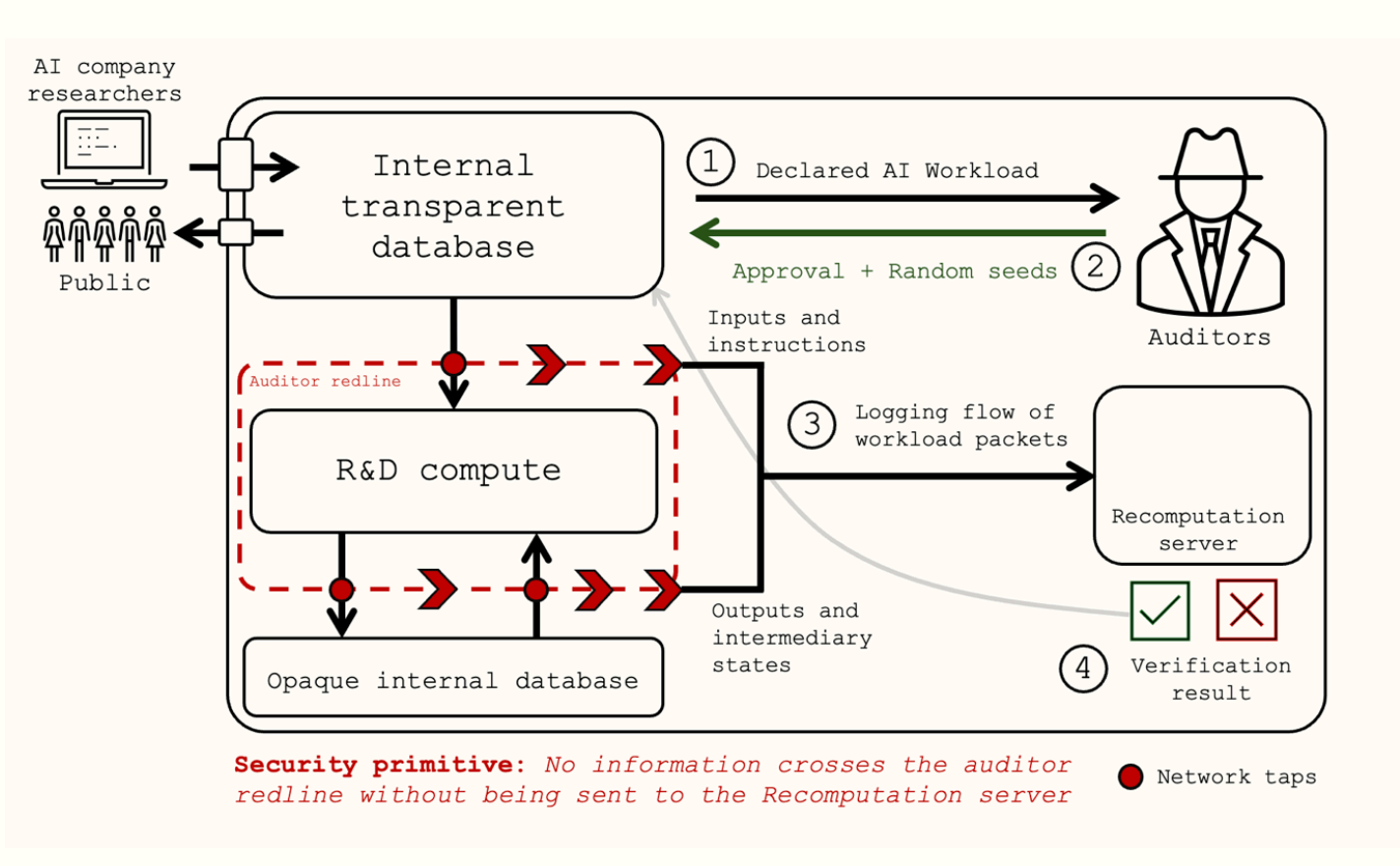
With a particular set of rules on research in place, there is a separate problem of enforcing these rules are actually followed. We can divide the problem of enforcing the R&D rules into two parts:

1. **Workload Approval**. Workloads that companies declare actually meet the regulatory rules, whatever their implementation may be.
2. **Workload Verification**. Workloads that the companies declare match what they actually run.

Our view is that workload verification is solvable in a similar way to what we discussed previously (evidence collection via network taps, and then partial recomputation of randomly selected reproducible packets), whereas workload approval will be closely tied to the form of AI R&D rules that are in place (e.g., are they just experiment compute caps? Are there specific banned techniques? Are there specific required techniques?) and our overall baseline proposal for approval is for this to just be manually carried out by teams of auditors in the R&D clusters.

▶ *Workload Approval.*

▶ *Workload Verification.*


AI company researchers
Public
Internal transparent database
① Declared AI Workload
② Approval + Random seeds
Auditors
Inputs and instructions
Auditor redline
R&D compute
③ Logging flow of workload packets
Recomputation server
Outputs and intermediary states
④ Verification result
Opaque internal database
Security primitive: No information crosses the auditor redline without being sent to the Recomputation server
Network taps

1. **Declaration**: AI companies declare the workload they want to run with full transparency on the code, algorithms, etc.
2. **Approval**: Auditors run checks per the **Workload Approval** implementation and if it passes they send back approval and provide necessary random seeds (removes a potential attack vector from AI companies gaming fake randomness).
3. **Evidence collection**: Auditors collect evidence from the workload (e.g., complete logging flow through network taps). In the diagram the evidence collection is depicted as low granularity, but in reality it could be happening at the level of individual servers, or GPUs or at even lower levels later on.
4. **Verification**: Auditors sample randomly and do partial recomputation and other checks to catch violations of workload correctness.
5. **Evaluations and release**: For workloads that consist of final training runs there can be an extra step that involves running evaluations that also get verified for correctness, and then final model approval for release leads to the model weights being downloaded from the opaque internal database and transported to the inference-only clusters. This process is extremely high stakes from a model weights theft perspective, so would need to be carefully designed and executed.

Late 2030: Secure inference-only clusters for post-deal model deployment come online

First SL5 inference-only clusters are online and start serving the first post-deal models that have been approved from the R&D clusters for broad public deployment. These clusters have the same basic verification approach as the retrofitted ones (network taps and partial recomputation) but with improved security properties. By the end of the year, around 40% of world compute (~370M of the global ~930M H100e) is in these new post-deal SL5 inference-only clusters.

Also by the end of 2030, the US and China have set up hardened cold storage facilities where they allow inspectors to verify each other's stash of AI chips, sized to be bigger than any plausible covert projects (a 5M H100e US bank and a 500K H100e China bank, per our deal decline supplement), and of relative size to each other similar to pre-deal status quo. These are in places that either side can easily defend. These locations will also be home to verified model weights stashes, so that models that the US and China agree would be positive sum to approve for use in the case of deal dissolution (e.g., models that would be very helpful for safety research and have strong safety cases). Around 2033 there is a model around the automated coder capability milestone that gets approved for the weights stash, and then in 2037 there is a near-frontier (top expert AI) model approved for the cold weights storage.

2031–2036+ Improving the deal

2031: Restrictions for hardware R&D and other high-risk domains

The core problem the US-China deal sets out to address is the intelligence explosion.[18] But the intelligence explosion isn't the only threat to deal with. Another core concern is that AI could enable an industrial explosion, too much AI hardware progress, or other dangerous scientific progress.

The case for being worried about AI software progress was illustrated in AI 2027: superhuman AI researchers could bootstrap to a superintelligence that we might not be able to control and whose goals we haven't figured out how to align. The case for being worried about too much hardware progress is superseded by this under a fast takeoff, but under the Plan A deal if we only slow down software progress a hardware driven takeover might actually be the majority of the takeover risk absent separate mitigations.

Three major reasons to be worried about unconstrained hardware progress:

> **Production explosion effects**
>
> **We should be worried about an explosion in the industrial capacity (e.g., number of robots) on earth, even absent design improvements.**
>
> With existing robot designs, it might become easy to rapidly build an absurd number of robots, by having new robots work on making more robots, naively the doubling times might be as low as months, which is very fast. The likelihood of an industrial explosion is argued in more detail in our economics supplement.
>
> An absurd number of robots might be able to do destabilizing things like:
>
> (1) Create massive military advantages for first movers (either directly in the form of a robot army, or through pure industrial capacity, e.g., missile/weapons production capacity),
>
> (2) Build AI hardware covertly (by building all the equipment & facility needed to do so from the ground up in a secret location)

> **Destabilizing robot designs**
>
> **We should also be worried about new advanced designs, e.g., self-replicating microbot/nanobot/biotech.**
>
> Biological anchors (and precedents of humans beating biology with intelligent designs) should have us worried that scary robot designs are possible. Specifically, we should be worried about self-replicating nanotech / biotech that have very fast doubling times.
>
> The core concern here is that these designs, unlike macroscale robots, might be too hard to govern, and break any regulations once they are invented.

> **Destabilizing AI compute designs**
>
> **We should also be worried about AI hardware design improvements that make it easier to manufacture.**
>
> AI hardware design improvements and/or paradigm shifts might make it easy to defect from the deal. If there is a paradigm shift in hardware design making it no longer require an extremely specialized, complex, and centralized manufacturing supply chain to make AI compute, it might be tractable for covert projects to manufacture more AI compute in an undetected way threatening the stability of the deal.

Similarly we want other high risk scientific domains such as biology research to have R&D restricted. Our default proposal for this is to rely on inference refusals and monitoring on the majority of public deployment, and then potentially set up dedicated locations that allow high-stakes hardware and other scientific research to be carried out in mutually verified research labs, where we make some (not necessarily identical) security-transparency tradeoff choices, and mutually verify that the companies follow the titration rules on hardware R&D, which can also be some combination of safety cases, ad-hoc rules, human interpretability requirements and resource caps (similar to software research).

▶ *Inference restrictions, monitoring and refusals for high-risk domains.*

▶ *Hardware research restrictions.*

2032: Cap & Trade for Robots and Compute

As mentioned earlier, we want to stop an explosion in the number of robots and AI chips on earth from threatening the stability of the deal. By default, our current naive modelling predicts that even with current hardware designs, the AI and robot capabilities that we forecast to be reached during the Plan A scenario would lead to a huge discrepancy between the raw cost of producing more AI chips and Robots and their economic value. By default, we think this leads to a reinvestment spiral and very fast doubling times.

In Plan A we propose a cap starting in 2032 that limits the number of robots and AI chips on earth to doubling times around 6 months, through cap & trade regulation, where private companies need to bid on the finite robot and AI chip production permits in any given time period. This cap should be set in units of usefulness of the robots for industrial production and AI compute in AI workloads, measuring both of these, especially across tasks and form factors in the case of robots, might be difficult and muddied (e.g., fuzzy lines between robots and more traditional machines). We nonetheless think it should be feasible to measure these things reasonably well for different robot and hardware designs, and allocate the cap & trade limits accordingly.

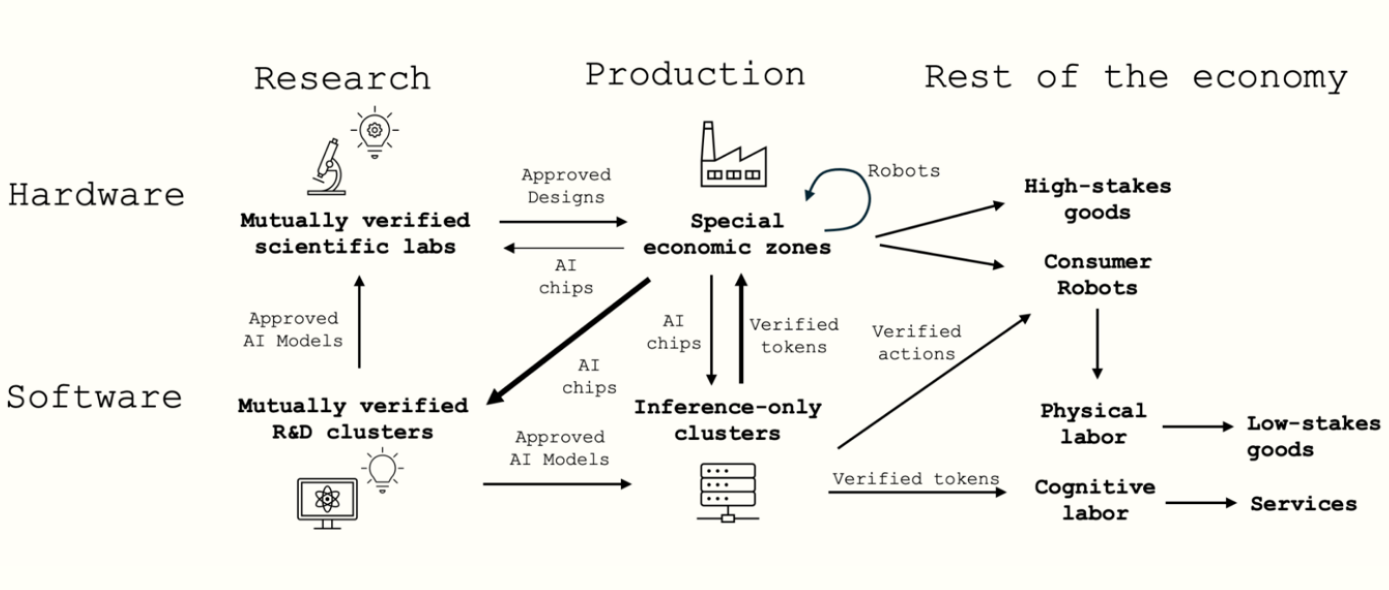
Actually enforcing the cap & trade permit rules might be very difficult if robot and/or AI compute production happens all over the world, so we propose centralizing new built up industrial capacity and robots in special economic zones (SEZs) to make for easier enforcement of the cap & trade rules.

▶ *How to enforce the cap & trade rules.*

The same cap and trade principle is extended to other products and services that are destabilizing in unrestricted quantities, but can be safe in lower quantities. Two major examples are products with edge compute and unmonitored inference. Things that might particularly rely on edge compute include self-driving cars and consumer compute, while unmonitored inference might be particularly valuable to governments for military operations. Under cap and trade the regime can accommodate these activities being legal, under safe societal-scale caps that maintain the stability of the deal. Then there can be markets for the permits the people, firms and the government can bid between each other to have the rights for.

▶ *Unverified edge compute cap.*

Putting together research titration and production capping, we have the following overview:



2033: Security improvements

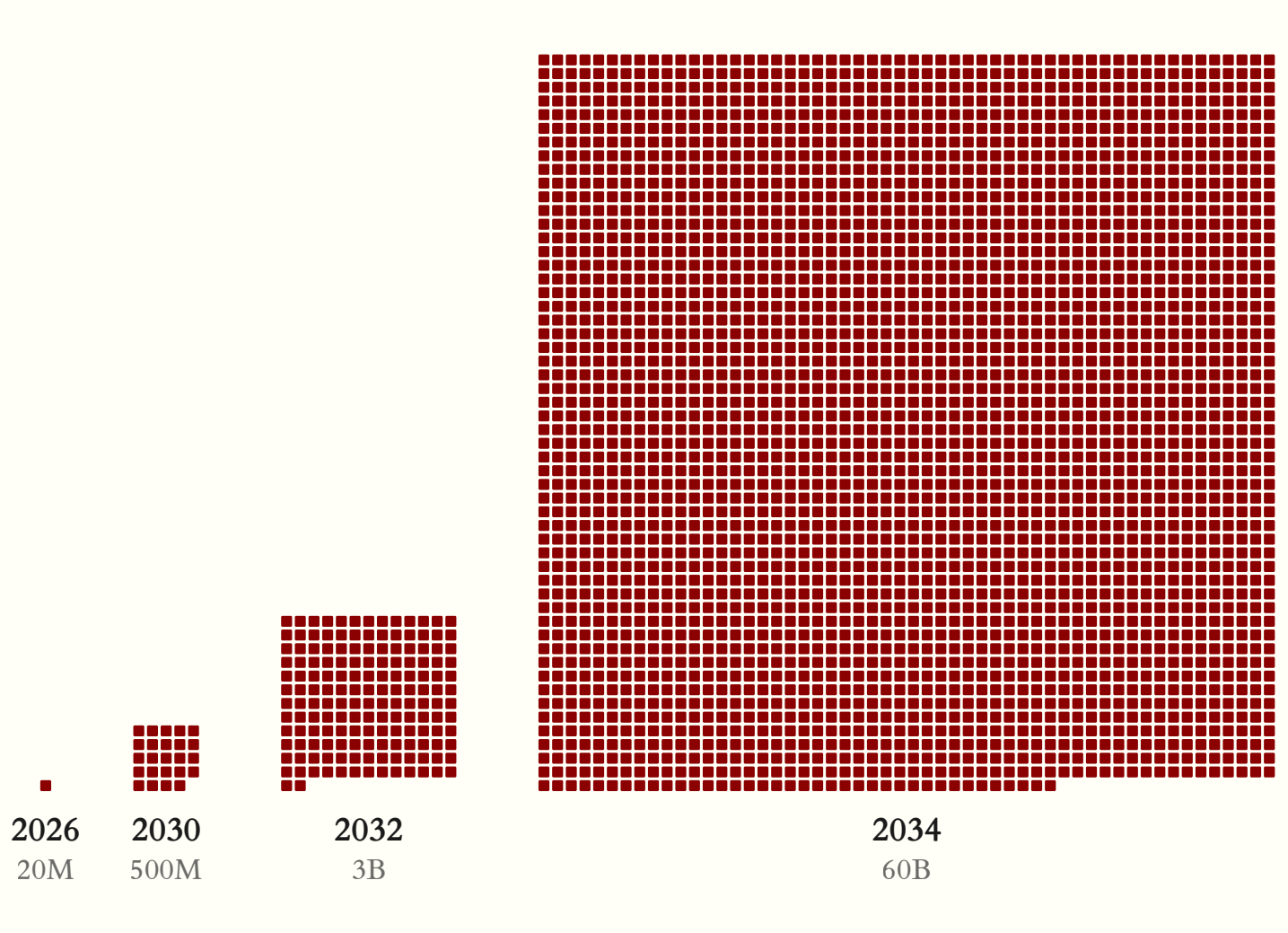
By 2033 we expect significant hardware security, and more general security properties of the datacenters, verification regime and capabilities to have improved significantly. By the end of 2033, around 99.9% of the worlds' compute is post-deal compute, meaning that it's trivial to just power down all pre-deal clusters and compute, and have the vast majority of compute be in new more secure locations built from scratch. Furthermore, by the end of 2034, around 90% of the compute that exists will have been produced during 2033 or 2034. This means that security improvements made by then can already be widely in place within just 1-2 years. Similar dynamics apply to the fabrication facilities.

Our uncertainty on what will be possible is high here, but some of the main ideas we have include:

- Datacenters built from scratch with extremely good side channel and airgapping properties.
- Huge progress in standardizing (for reproducibility) and improving security of software stacks for inference and R&D workloads more generally.
- New compute production facilities (fabs) built from scratch with mutual monitoring measures and being colocated with supply chain and potentially even with final centralized datacenters they will fill to minimize attack surface.
- The beginning of some hardware-software co-dependencies especially with architecture changes or number formats, so that optimization and progress in these paradigms is less applicable on pre-deal compute
- Increases to the logging / evidence collection granularity of the verification regime, especially of the R&D datacenters, with logging technology being compatible with extremely high bandwidth data paths on backend networking to reduce overhead.
- Generally improved hardware security properties with more robust trusted execution, secure boot, encryption, reproducibility, etc., which are especially important for recomputation servers. Some degree of standardization in hardware may also be helpful for reproducibility.
- Special hardware designs for automatic on-device verification of certain workloads, in particular, hardcoded weights on some chips to verify they're inference-only. This might be particularly useful for 'edge verification' allowing for more AI compute in consumer products without risk of use in covert projects.

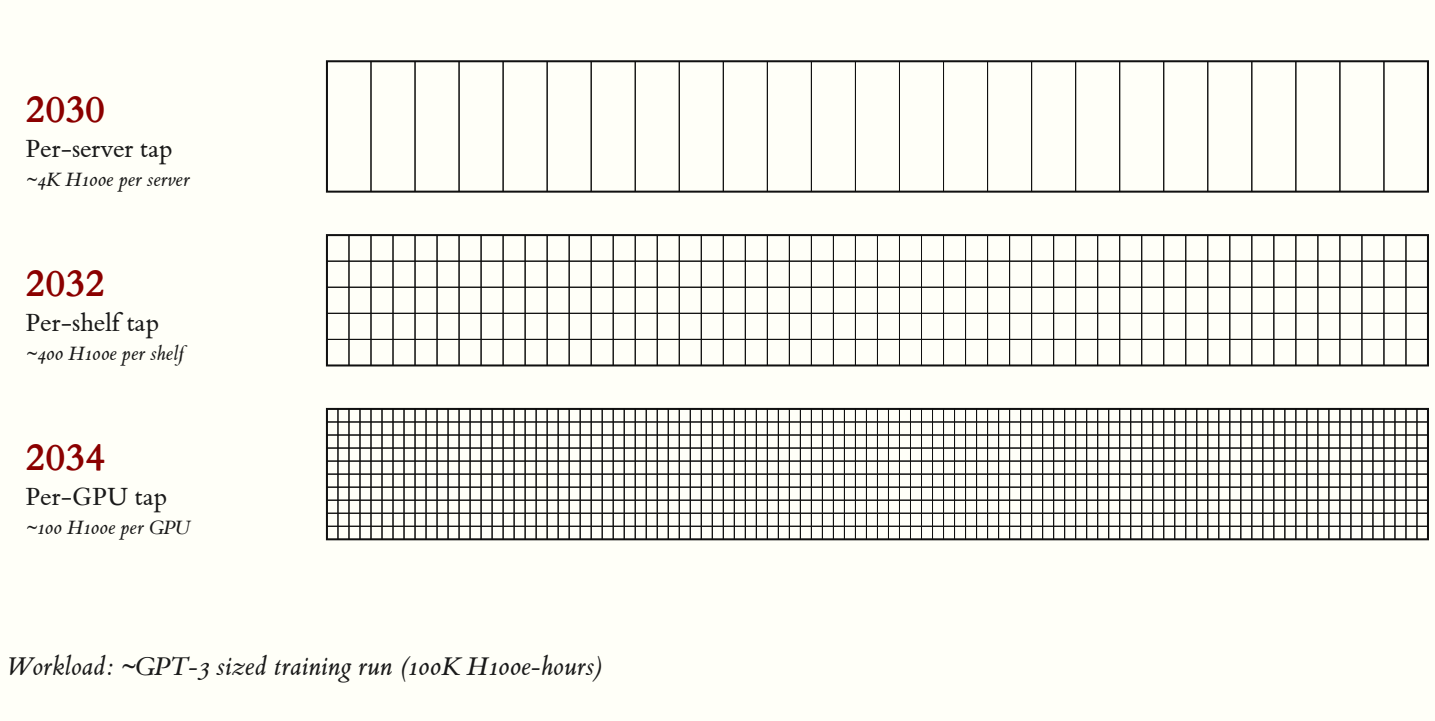
2034: Verification improvements

Through the 2030s the total amount of AI compute in the world grows explosively. One important dynamic to note is that the verification burden (the need to have higher coverage) grows as the total compute under verification grows. In Plan A, we both forecast and recommend a significant hardware buildup (capped by the cap & trade regime to stay verifiable) for other reasons related to tradeoffs with deal stability (hardware is less likely to leak to illegal projects so differentially increases the legal projects affordances).



World Compute – `ai-2040.com`

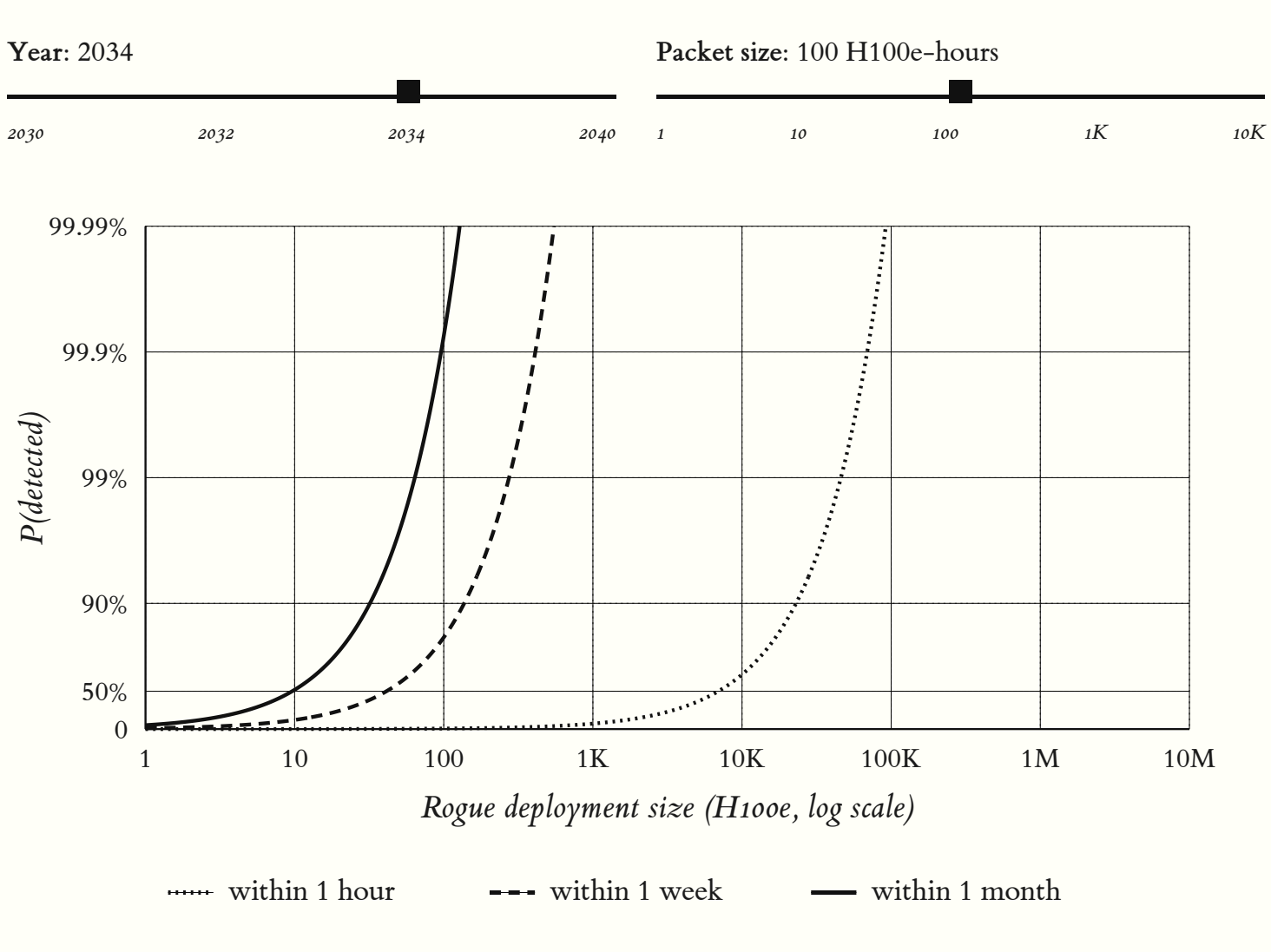
The verification assurance curve needs to improve such that any potential nonnegligible rogue internal deployment would have a high chance of detection. By 2034, with one packet collected per hour from each frontier GPU (~100 H100e each, so ~100 H100e-hour packets) and a 1% recomputation budget, the appendix math shows the verifier catches any rogue internal deployment at 99% confidence after it accumulates ~46,000 H100e-hours of unapproved work, so a sustained 10K H100e deployment is detected within ~4.6 hours, a 1K within ~46 hours, and so on.



Verification logging granularity – `ai-2040.com`

This is an ambitious verification scale, and it also relies on extremely good integrity security on the verification regime. We think it will be possible to scale it this far, especially with AI assistance, but are not confident, and find it plausible that the compute cap should be set lower in order to not increase the verification burden this much.

Pool: ~33B H100e (2034 buildout)
Recomputation budget: 1%



Catching a rogue internal deployment – `ai-2040.com`

2035: Moving the datacenters to the ocean

One of the biggest ongoing risks is that the deal breaks down and actors return to racing. Because of the vast increase in compute scale that has not been turned into capabilities progress (due to the safety regulations) the return to racing could be even more dangerous than the initial status quo, despite the alignment progress made up until this point, because things would now go faster. This motivates having measures in place to return to pre-deal compute levels, or even smaller compute scales if the deal breaks down.

We think the relative engineering and logistical costs involved in building new datacenter capacity on the ocean in international waters will be low enough due to robot and AI labor to make this worth it around 2035, because it would become much harder for any particular actor to defend and much less escalatory to destroy in the event of the deal breaking down. This is an intentional measure for mutually assured compute destruction to increase the deal's stability.

2036+ Advanced robustness improvements

Later into Plan A we are increasingly uncertain about what robustness improvements will be possible in terms of security, verification and deal stability. Potentially extremely ambitious things will be possible, a brainstorm here included:

- Completely rewritten software stacks to be significantly or entirely formally verified.
- Huge red teaming efforts on the verification scheme or security measures using resources that dominate any potential covert actors.
- Using AI and robot labor to track down potential covert projects in mutually agreed, symmetrical ways, e.g., the 'AI silver bullets' explored in our covert project supplement: reliable lie detectors, privacy-preserving AI verification, and extreme industrial verification.

Appendix: Packet-Based Verification

This appendix works through the math on random sampling for use in partial recomputation to show why it scales so favourably for detecting incorrect workloads. A key simplifying assumption we make in this analysis is that the workload packets are perfectly reproducible.

A.1 Setup

In this analysis we treat verification as a sampling problem on a stream of packets. A packet is a reproducible unit of computation that the verifier can recompute: its inputs, instructions, and outputs are all collected at the evidence-collection point (e.g., a network tap). What the computation actually is (one forward pass, one training batch) doesn't affect the math below. Further simplifying assumptions we make are:

1. **Uniform size.** Every packet represents the same amount of underlying compute. Non-uniform packets work too, with a weighted-sampling argument that doesn't change the qualitative results.
2. **Independent random sampling.** The verifier audits each packet independently with the same probability. In reality you would sample without replacement.

Let the verifier's pool be some collection of packets; it could be a single training run, every packet from every workload running under the deal in a given month, or one cluster's traffic. Let N_{packets} be the total number of packets in that pool, $s = 1/N_{\text{packets}}$ the packet size (the fraction of the pool each packet represents), C the recomputation budget (the fraction of the pool the verifier can afford to recompute), and $N_{\text{verified}} = C \cdot N_{\text{packets}} = C/s$ the number of packets actually audited.[20]

A.2 Detection by Random Sampling

The detection math in this section was independently derived but matches the analysis in Rinberg et al. (2025). Suppose an attacker substitutes N_{fake} packets, replacing the declared computation in each with their own unauthorized one. Let

$$F_{\text{fake}} = \frac{N_{\text{fake}}}{N_{\text{packets}}}$$

be the fraction of packets that are fake. If the verifier audits each packet independently with probability C, the exact probability that none of the N_{fake} fakes is audited is

$$P(\text{undetected}) \;=\; (1-C)^{N_{\text{fake}}}.$$

For small C and modestly large N_{fake}, this is well approximated by a Poisson distribution on the number of caught fakes, with mean $\lambda = N_{\text{verified}} \cdot F_{\text{fake}} = C \cdot N_{\text{fake}}$:

$$P(\text{undetected}) \;\approx\; e^{-N_{\text{verified}} \cdot F_{\text{fake}}},$$

so the probability of catching the attack (the **detection confidence** at attack size F_{fake}) is

$$P(\text{detected}) \;=\; 1 - P(\text{undetected}) \;\approx\; 1 - e^{-N_{\text{verified}} \cdot F_{\text{fake}}}.$$

We'll use the Poisson form throughout, since it's the cleanest expression of the scaling. It breaks down only when C approaches 1 or N_{fake} is single-digit, which is also when random sampling stops buying you anything (see the discussion at the end of §A.3).

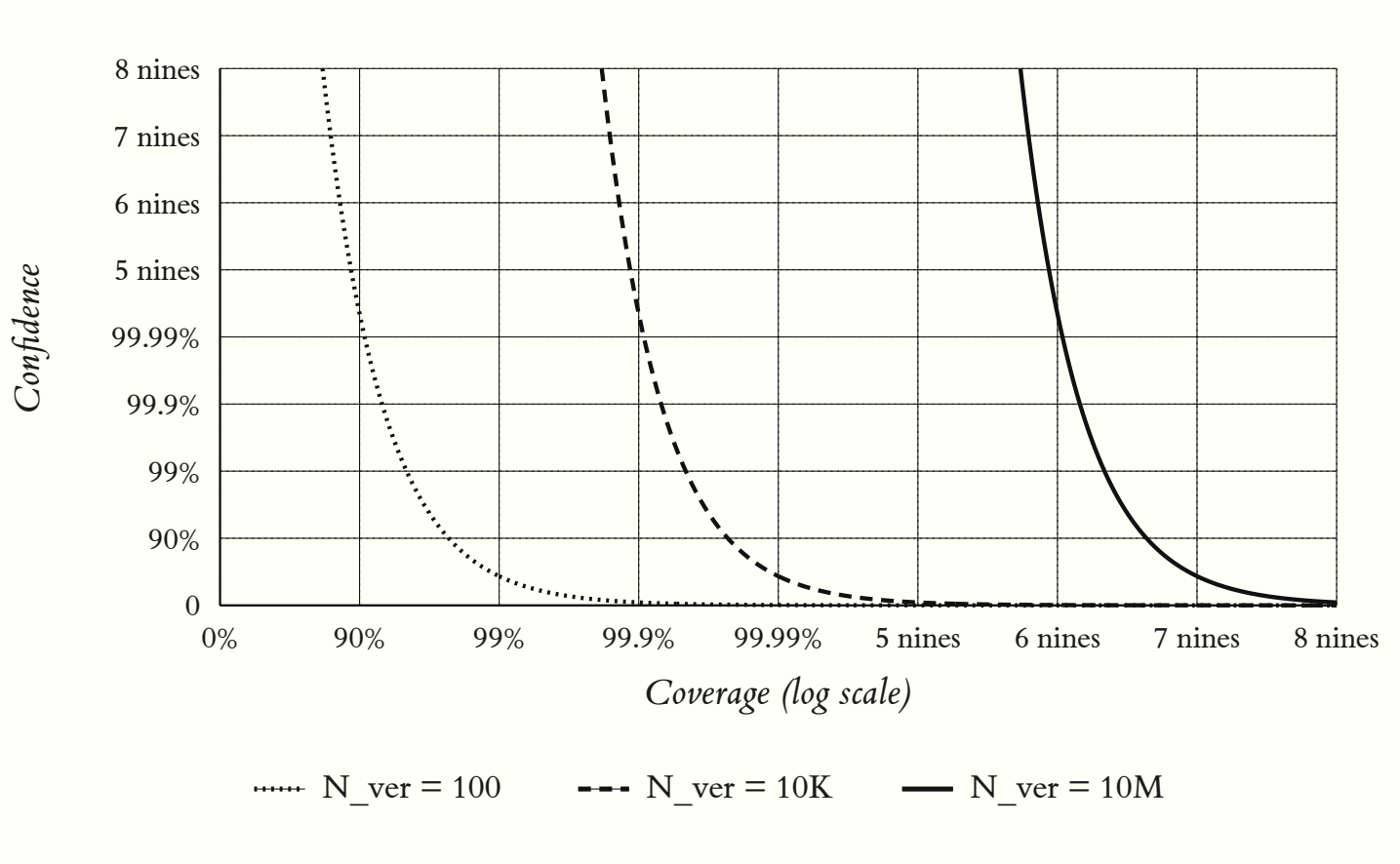
A.3 The Assurance Curve

The assurance curve is determined by **coverage** and **confidence** which can be expressed in terms of F^*_{fake}, the largest fraction of fake packets we're willing to tolerate without detection:

- **Coverage** $= 1 - F^*_{\text{fake}}$, the fraction of the workload we're claiming is legitimate.
- **Confidence** $= 1 - e^{-N_{\text{verified}} \cdot F^*_{\text{fake}}}$, the probability we'd catch any actual violation at or above that threshold.

The confidence formula comes from §A.2: $P(\text{detected}) = 1 - e^{-N_{\text{verified}} \cdot F_{\text{fake}}}$, evaluated at the threshold F^*_{fake}. The recomputation budget shows up through $N_{\text{verified}} = C \cdot N_{\text{packets}}$, so more audited packets push the curve outward.

Plotting confidence against coverage for several verification budgets (i.e., different values of N_{verified}):



Assurance curves – `ai-2040.com`

These curves show the confidence-coverage relationship at different values of N_{verified}. Higher N_{verified} gives you higher assurance (shifts the curve right). Since $N_{\text{verified}} = C/s$, **the curve can be shifted higher either by (1) increasing the recomputation budget C or (2) shrinking the packet size s.**

▶ *When the Poisson approximation breaks down.*

A.4 Worked Examples

A.4.1 100K-token Backdoor in a 100T-token Training Run

Consider a training run of $N_{\text{tokens}} = 10^{14}$ tokens (100T) using distributed data parallelism. Network taps between server racks log every gradient step (its input batch, the gradient update, and the resulting checkpoint update), so each gradient step is one packet. An attacker wants to insert a backdoor by silently training on $N_{\text{fake tokens}} = 10^5$ off-spec tokens (100K).

Quantity	Symbol	Value
Training tokens	N_{tokens}	10^{14}
Attack size (backdoor)	$N_{\text{fake tokens}}$	10^{5}
Batch size	B	design choice
Number of packets	N_{packets}	N_{tokens}/B
Packet size	s	B/N_{tokens}
Fake fraction	F_{fake}	$N_{\text{fake tokens}}/N_{\text{tokens}} = 10^{-9}$

For 99.9% detection confidence we need $N_{\text{verified}} \cdot F_{\text{fake}} \geq \ln(1000) \approx 7$, i.e. $N_{\text{verified}} \geq 7 \times 10^9$. The recomputation cost as a fraction of the training run is then

$$C \;=\; \frac{N_{\text{verified}}}{} \;=\; N_{\text{verified}} \cdot s \;=\; \frac{7 \cdot B}{} \;=\; \frac{7B}{}$$

---

[17] This is explored more in our capabilities scaling strategy supplement. There are also a few deeper layers of nuance here that complicate the picture, including what transparency regime is in place as discussed in our transparency supplement.

[18] An R&D efficiency spiral in AI research, where the feedback loop of smarter AIs → better research → smarter AI, holds through to AIs that we cannot control or align, and slowing down to invest in control/alignment is uncompetitive, locking companies and countries into a multipolar trap. Coordination, that is verified, is the path out.

[20] Note none of these quantities depend on how the work is divided into individual workloads. What matters is the pool the verifier samples from. If there are 100 different training runs going on simultaneously and the verifier samples uniformly at random across all their packets, then 'the workload' in the math below is the union of all 100. The per-workload framing in the worked example (§A.4.1) is a convenience for talking about one concrete attack.

$$N_{\text{packets}} \qquad N_{\text{verified}} \qquad N_{\text{fake tokens}} \qquad 10^5.$$

Batch size B	N_{packets}	Recomputation cost C
10,000	10^{10}	70%
1,000	10^{11}	7%
100	10^{12}	0.7%
10	10^{13}	0.07%

Making batches 10× smaller cuts the required recomputation cost 10× for the same confidence (ignoring the fact that we do better than this at values of C close to 1 by sampling without replacement). The two practical levers on packet size are (1) smaller batches (more gradient steps per training run), and (2) sub-batch logging, tracking intermediate computations within a batch (e.g., individual forward/backward passes), which might require evidence-collection devices inside server racks or even inside GPUs themselves.

A.4.2 Bounding Rogue Compute Use in 100B H100e of Verified R&D

The 2034: Verification improvements section explores what **rogue internal deployments** have low detection probability if the verifier has a 1% recomputation budget across 100B H100e of verified R&D compute. Rogue internal deployments are potentially scarier than external illegal projects due to internal access to frontier models and the rest of the R&D cluster. Detection relies on the verification system catching their fake packets. The math below translates the assurance curve from §A.3 to concrete values.

Starting from $P(\text{detected}) = 1 - e^{-N_{\text{verified}} \cdot F_{\text{fake}}}$ and substituting $N_{\text{verified}} = C \cdot N_{\text{packets}}$ and $F_{\text{fake}} = N_{\text{fake}}/N_{\text{packets}}$, the N_{packets} cancels and the detection probability becomes:

$$P(\text{detected}) \;=\; 1 - e^{-C \cdot N_{\text{fake}}}.$$

In this form the detection probability is a function of the recomputation budget C and the absolute number of fake packets N_{fake} the attacker generates.

For 99% confidence we need $C \cdot N_{\text{fake}} \geq \ln(100) \approx 4.6$, so the attacker is caught with 99% confidence the moment they have accumulated

$$N_{\text{fake}}^{99\%} \;=\; \frac{4.6}{C}$$

fake packets. With $C = 1\%$ that is 461 packets, independent of pool size and of how the work is spread across workloads. Multiplying by the packet size p converts this to a bound on cumulative rogue compute work:

$$W_{\text{rogue}}^{99\%} \;=\; \frac{4.6 \cdot p}{C}.$$

Consider the operational setup of the 2034 section, in which the verifier collects one packet per hour from each frontier GPU (about 100 H100e each by 2034). Then $p = 100$ H100e-hours, and with $C = 1\%$,

$$W_{\text{rogue}}^{99\%} \;=\; \frac{4.6 \cdot 100}{0.01} \;=\; 46{,}000 \text{ H100e-hours}.$$

The verifier catches with 99% confidence any rogue internal deployment that accumulates more than about 46K H100e-hours of unapproved compute work inside the verified pool, regardless of pace or distribution. The bound on cumulative rogue work scales linearly in p/C; a few representative operating points:

Packet size p	Budget C	Max undetected rogue work
100 H100e-hours	1%	46,000 H100e-hours
100 H100e-hours	0.1%	460,000 H100e-hours
10 H100e-hours	1%	4,600 H100e-hours
1 H100e-hour	1%	460 H100e-hours